
BDG: Batch-Dispersion Guidance for Property-Band Generation with Continuous Flow Matching

Bobo Li,¹ Henry Huang,¹ Idea Idehpour,¹ Haimo Fang¹

¹Department of Computer and Information Science, University of Pennsylvania

Abstract

A design campaign accepts every sample inside a tolerance band around a target, and we study what limits that fraction under inference-time guidance. On QM9 3D coordinates we train an E(3)-equivariant flow-matching model, compare it against a diffusion model, and confirm guidance works, roughly doubling coverage while molecule stability falls from 0.434 to 0.246. The property error splits into centring and spread: at a median target centring is small relative to the band while spread is several band widths, and one strength dial moves both. Batch-Dispersion Guidance measures the batch spread of the guide’s endpoint prediction, forms a scale-free error against a requested setpoint, and servo the guidance term’s deviation gain by it while holding the centring gain fixed, at no additional network evaluations. The setpoint moves achieved spread monotonically on six of six property curves, from 0.67 to 1.22 times unguided, including widening, which the strength dial does not reach. Coverage does not improve over a tuned plug-in baseline: the coefficient is affine in the predicted property, so the rule is the plug-in field reweighted. Transferred unchanged to the probability simplex, every signature reproduces at a smaller measured effect. The controller buys calibrated spread and chemical fidelity at matched coverage.

1 Introduction

Continuous generative models learn a transport from a simple reference distribution to a complex one. Diffusion reverses a stochastic corruption [1] and flow matching regresses a vector field along a prescribed path [2]; stochastic interpolants contain both [3], so one backbone serves either and the choice is empirical. Can a frozen property predictor steer a pretrained generator?

A design campaign accepts any molecule inside a tolerance band. At a median target our batches sit one or two band widths from the target while their spread is five to nine band widths, so clustering limits coverage. Every guidance rule exposes one strength dial moving centring and spread together [4, 5]; recent work makes that scale state dependent [6, 7] and couples trajectories repulsively [8], and none lets the user request a spread in property units.

We ask one question on two state spaces: does making batch dispersion a requested quantity raise band coverage? Modality 1 is QM9 3D coordinates and Modality 2 is DNA enhancers on the probability simplex, and we expect a calibrated spread knob to raise coverage and to transfer, since the controller reads a scalar variance carrying no geometry.

- We compare flow matching and diffusion under one split, evaluator and budget.
- We demonstrate guidance against a matched unguided control and price it.
- We introduce BDG, servoing the deviation gain by the batch variance of the guide’s own endpoint prediction at no extra network evaluations, and state the reduction bounding it.
- We transfer the controller to the simplex and test whether it generalizes.

2 Related Work

Continuous models and guidance. Flow matching [2], rectified flow [9] and EDM [1] are endpoints of one construction [3, 10]. Our five comparators are DPS [11], LGD-MC [12], TMPD [13], TFG [14] and D-Flow [15].

Near our mechanism. Prior work drives ensemble moments to a target [16], tilts or aligns ensemble variance [17, 18], couples trajectories repulsively [8], widens through negative guidance [19], and reads the guidance scale as a control gain [7, 6]. Over roughly eight queries we did not find a weight set by the batch variance of a *learned predictor’s* endpoint prediction against a *two-sided user-chosen* setpoint.

The two modalities. EDM defines the stability conventions we adopt [20]; see also [21–24]. On the simplex, Dirichlet [25], Fisher [26] and Gumbel-Softmax [27] flow matching and MOG-DFM [28] define the task and score it by Fréchet biological distance at 500 base pairs.

3 Methods

3.1 Modalities, Data, and Preprocessing

Modality 1 is QM9, 133,885 molecules on $\mathbb{R}^{N \times 3}$: at most 29 atoms, coordinates zero-centred, a hard one-hot over {H,C,N,O,F}, no feature scaling. Splits are `train_a` (51,527) for the generator and guide f_A , `train_b` (51,527) for the evaluator f_B , `val` (17,748) for selection, `test` (13,083) held out, so scoring is independent of the guide. Generation is unconditional, and the properties are dipole, polarizability and gap.

Modality 2 is 6,126 *Drosophila* Kenyon-cell enhancer regions on $(\Delta^3)^L$, decoded by per-position `arg max`, a state space with no rotation group and an exact property.

3.2 Continuous Flow Matching Baseline

With $t : 0 \rightarrow 1$ from noise to data, the linear interpolant $x_t = tx_1 + (1-t)x_0$ has constant conditional velocity $u_t = x_1 - x_0$, and the source is projected onto the zero-centre-of-mass subspace of dimension $3(N-1)$. The objective is a masked mean over $t \sim \mathcal{U}(0,1)$, coordinates and types normalised separately, $\lambda = 1$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,x_0,x_1} \left[\frac{1}{3\Sigma M} \|M \odot (v_\theta^c - u^c)\|^2 + \lambda \frac{1}{K\Sigma M} \|M \odot (v_\theta^f - u^f)\|^2 \right], \quad (1)$$

with M the atom mask and K the type channels. The network is a dense $E(n)$ -equivariant graph network, width 256, 8 layers, 3.75 M parameters, Adam at 2×10^{-4} , batch 256, EMA 0.9999, 1500 epochs (303k steps), selected on validation atom stability. Sampling is explicit Euler, 100 steps.

3.3 Diffusion Baseline

The diffusion baseline shares everything with Section 3.2 but the corruption, which makes Section 4.2 interpretable: a variance-preserving process with $\beta(\tau) = \beta_{\min} + \tau(\beta_{\max} - \beta_{\min})$, $\beta_{\min} = 0.1$, $\beta_{\max} = 20$:

$$x_\tau = \alpha(\tau)x_0 + \sigma(\tau)\epsilon, \quad \log \alpha(\tau) = -\frac{1}{2}(\beta_{\min}\tau + \frac{1}{2}(\beta_{\max} - \beta_{\min})\tau^2), \quad \alpha^2 + \sigma^2 = 1. \quad (2)$$

The network predicts ϵ . We integrate $dx/d\tau = -\frac{1}{2}\beta(\tau)(x + s_\theta)$ backwards, $s_\theta = -\epsilon_\theta/\sigma$, on a decreasing 100-step grid, then jump to the Tweedie mean: budget 101 against the flow model’s 100.

3.4 Guided Generation

Each guided step toward target y forms the endpoint estimate m_i , which is $x_t^{(i)} + (1-t)v_\theta(x_t^{(i)}, t)$ for the flow model and the Tweedie mean for the diffusion model, evaluates $F_i = f_A(m_i)$ and $g_i = \nabla_m f_A(m_i)$, and adds

$$G_i = \frac{\text{num}_i}{s^2} J_i^\top g_i, \quad \text{num}_i^{\text{plug}} = y - F_i, \quad (3)$$

with $J_i^\top$ the vector-Jacobian product and s the guide’s target standard deviation. The sampler scales this by $w(1-t)/t$ and applies a velocity-relative trust region: removing it produces 222 non-finite samples against none clipped. Guidance runs for $t \geq 0.5$.

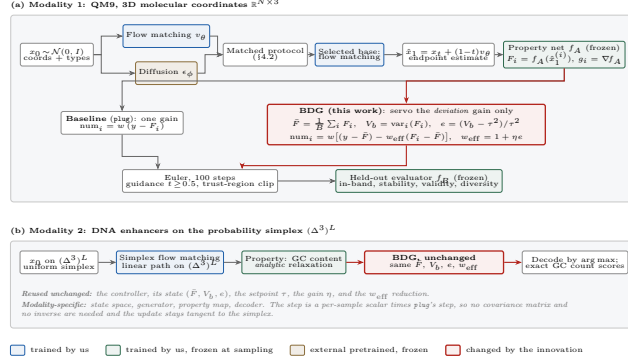


Figure 1: Study overview. **(a)** Modality 1: one representation feeds a flow-matching and a diffusion model; the plug-in baseline applies one gain to $(y - F_i)$, BDG holds the centring gain at w and serves only w_{eff} . **(b)** Modality 2: the controller transfers unchanged. Colour marks trained, frozen, pretrained, and changed-by-the-innovation components.

3.5 Proposed Innovation: Batch-Dispersion Guidance

Mechanism. For a batch BDG forms the mean $\bar{F}$, the unbiased variance V_b and the scale-free error $e = (V_b - \tau^2)/\tau^2$ against a setpoint τ , then replaces the numerator of Equation (3) by

$$\text{num}_i = (y - F_i) - \eta e (F_i - \bar{F}) = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}), \quad w_{\text{eff}} = 1 + \eta e. \quad (4)$$

$V_b > \tau^2$ contracts the batch and $V_b < \tau^2$ expands it; both terms multiply the same g_i , so BDG costs what plug-in costs. Algorithm 1 is the whole step and Figure 2 is it measured.

The reduction. Equation (4) is affine in F_i , so it factors exactly,

$$(y - F_i) - \eta e (F_i - \bar{F}) = (1 + \eta e) (y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{1 + \eta e}, \quad (5)$$

verified to 3.0×10^{-15} analytically and 6.5×10^{-7} on the real field. BDG is therefore the plug-in field at a rescaled weight aiming at a shifted target.

Properties and controls. Since $V_b \geq 0$, $w_{\text{eff}} \geq 1 - \eta$ for any batch, licensing two-sided operation and forcing $\eta > 1$. A per-sample posterior variance with $k = (1 - t)^2/t$ fires its expansion branch unconditionally and must be clamped. The equilibrium $V^* = \tau^2(1 - 1/\eta)$ is an asymptote the 50-step window does not reach, so we never describe τ as a value the sampler attains. Two settings are *gates*, not controls: $\eta = 0$ and the one-sided variant must each reproduce the plug-in field bit-identically. Three controls isolate the mechanism: a strength sweep at fixed (η, τ) , an open-loop replay of a recorded e schedule on another seed, and a matched signed- w plug-in control, under a drop rule: if it reaches BDG’s spread at equal chemistry, the claim is withdrawn.

3.6 Transfer to Modality 2

BDG’s step is a per-sample scalar times the plug-in step, so it needs no inverse, which matters where $\text{diag}(p) - pp^\top$ is singular. What changes is the state space, the generator and the property map, an analytic relaxation of GC content whose exact count is the evaluator.

3.7 Model Overview

Figure 1 shows the whole study: the two base models, the guidance signal, the component BDG changes, and what transfers to Modality 2.

4 Results

4.1 Evaluation Protocol

The target metric is IN-BAND, the fraction of molecules whose property, scored by the held-out f_B , lies within $\pm\delta$ of the target, with $\delta = 2 \times$ the evaluator’s calibration mean absolute error on 3,000

Table 1: Base models on QM9, unconditional, 100-step Euler probability-flow ODE, our evaluator under EDM conventions. Our flow row is three seeds of 10,000 samples; EDMsecond is borrowed, $n=2,000$, one seed, so it is not comparable. — marks a value the v3 run has not produced (Appendix A.6).

Model	Source	Atom stab.↑	Mol. stab.↑	Validity↑	Uniq.↑	NFE↓	Params
Flow matching (ours)	reproduced	0.9366	0.3993	0.7625	0.9939	100	3.75 M
VP diffusion (ours)	reproduced	—	—	—	—	101	3.75 M
EDMsecond [20]	borrowed	0.9644	0.6725	0.8490	—	100	5.34 M
EDM [20]	quoted	0.9870	0.8200	0.9190	—	1000	—
Real QM9 (ceiling)	measured	0.9940	0.9560	0.9820	1.0000	—	—

Table 2: Guided against unguided at q50, otherwise matched. One generator, one seed, $n=256$, best strength per arm, no chemistry floor. Cells are μ / gap / α .

Arm	IN-BAND↑	Spread/unguided	Mol. stab.↑
unguided	0.090/0.121/0.051	1.00/1.00/1.00	0.434/0.434/0.434
plug-in (best w)	0.184/0.219/ 0.133	0.52/0.60/0.47	0.246/0.199/0.320
TFG ($w=4$, μ only)	0.535 /—/—	0.22/—/—	0.172/—/—
BDG ($w=32$)	0.180/ 0.227 /0.106	0.49/0.68/0.53	0.348 /0.262/0.340

Table 3: BDG ablations at q50, ordered by *measured* w_{eff} . Rows above the rule are gates, below are controls. Port cells: $n=256$, one seed, $w=1$; the three-seed v3 grid is in Appendix A.3. [v3 ablation stage]

Variant	Role	η	τ_{mult}	Spread/ung.	IN-BAND↑	Mol. stab.↑
plug-in, $w=1$	base	—	—	0.714	0.1094	0.3633
BDG $\eta=0$	gate	0	1.0	0.714	0.1094	0.3633
BDG one-sided, expand	gate	4	1.5	0.714	0.1094	0.3633
BDG contract	ladder	4	0.5	0.670	0.1406	0.3359
BDG neutral	ladder	4	1.0	0.900	0.0820	0.4258
BDG expand	ladder	4	1.5	1.093	0.0703	0.4141
plug-in, $w=4$	control	—	—	0.584	0.1406	0.2656
open-loop e replay	control	4	0.5	0.686	—	—
signed- w plug-in	control	—	—	—	—	—

disjoint molecules. We report IN-BAND decoded, the decomposition $\text{RMSE}^2 = \text{bias}^2 + \text{spread}^2$ in units of δ , and the spread ratio against unguided, all at the **q50** target. Arms share the checkpoint, guide, evaluator, target, atom counts and noise, paired within a backend at $w=1$; equal w is not equal force, the applied-correction share spanning $5.8\times$. The headline uses three seeds.

4.2 Modality 1: Flow Matching versus Diffusion

In Table 1 our flow model reaches 0.9366 atom and 0.3993 molecule stability at 100 evaluations, saturating by 500 steps. Ranking the causes of the gap to published equivariant diffusion: unscaled one-hot features lead, the published ablation losing 3 atom and 35 molecule points; then budget, 303k steps against roughly 1.56M; the half-data split is ruled out at 0.33 atom points, and the evaluator by bond tables matching the reference. We carry flow matching forward without a fidelity win, since the matched diffusion row is pending, and its linear path supplies the endpoint estimate. [restate when v3 Job A lands]

4.3 Modality 1: Guidance Works

In Table 2 guidance raises IN-BAND by $2.0\times$ on μ , $1.9\times$ on the gap and $2.6\times$ on α for the plug-in rule, and TFG reaches $6.0\times$ on μ . Molecule stability falls from 0.434 to 0.246 for the plug-in rule and to 0.172 for TFG. Spread contracts to between 0.22 and 0.60: the arm that contracts most wins IN-BAND and loses the most chemistry.

4.4 Modality 1: Innovation Ablations

The gates pass exactly in Table 3. At $\eta=0$ BDG is bit-identical to the plug-in rule; the one-sided variant likewise, its raw pre-clamp error of -0.984 clamped to 0. **The setpoint moves spread monotonically** on six of six curves, 0.670 to 1.219 times unguided; the deviation gain changes sign up to 34 times and spans $[-2.64, +10.35]$ against the bound $w_{\text{eff}} \geq -3$. **Coverage does not improve.** No rung beats the plug-in rule at $|z| \geq 3$: over 72 paired tests the largest is 3.07, a Šidák-corrected $p=0.14$. **The controls.** Plug-in at $w=4$ matches BDG’s contraction rung at 0.1406 IN-BAND on

Table 4: Five recent inference-time guidance methods, all reproduced on our frozen checkpoint, guide and evaluator at $q50$ and $w=1$. Passes are per cell, $n=5,000$.

Method	Status	IN-BAND $\uparrow$	Mol. stab. $\uparrow$	Diversity $\uparrow$	Gen. passes	Guide passes
DPS-style plug-in [11]	reproduced	—	—	—	10,000	4,000
TMPD-inspired [13]	reproduced	—	—	—	10,000	4,000
LGD-MC [12]	reproduced	—	—	—	12,000	20,000
TFG [14]	reproduced	—	—	—	8,000	20,000
D-Flow [15]	reproduced	—	—	—	—	—
Flow matching, unguided	internal	—	—	—	10,000	0
BDG (ours)	reproduced	—	—	—	10,000	4,000

Table 5: Transfer to DNA enhancers on the simplex. Signatures are exact statements about the controller. Our base generator is the linear flow-matching model these papers baseline against, so their numbers are never pooled with ours.

Signature	QM9	DNA simplex
$\eta=0$ reproduces plug-in	bit-identical	bit-identical
one-sided at an expanding setpoint	exact	exact, $w_{\text{eff}}=+1.000$
setpoint ladder monotone in spread	$0.67 \rightarrow 1.22$	$0.81 \rightarrow 1.01$
cost identical to plug-in	yes	yes, byte-identical
contraction breaks the fidelity floor	yes	yes
IN-BAND improves	no	no
<i>Against recent simplex sequence models</i>		
Dirichlet FM [25]		[v3 Job B]
Fisher Flow [26]		[v3 Job B]
MOG-DFM [28]		[v3 Job B]
BDG, transferred (ours)		[v3 Job B]

0.2656 molecule stability against BDG’s 0.3359; the open-loop replay recovers 69.5% to 96% of the closed loop against plug-in’s 8.4% to 30.5% gap.

4.5 Modality 1: Comparison with Recent Methods

Table 4 compares guidance rules on one frozen generator. Appendix A.4 records three port caveats. BDG uses 4,000 guide passes per cell against 20,000 for LGD-MC and TFG. [better/comparable/worse discussion]

4.6 Modality 2: Transfer of the Innovation

In Table 5 BDG transferred with no change to the controller, and coverage does not improve. The effect is $2.6\times$ smaller, 0.20 against 0.52: GC is affine in the simplex coordinates, it averages over 200 positions, and the band spans only 4.3 attainable values at granularity $1/200$.

4.7 Cross-Modality Analysis and Failure Modes

The direction matches across modalities and the magnitude does not, which the property map explains. Two measured confounds bound the result. The trade-off is re-parameterised and not removed: chemistry cost tracks $|w_{\text{eff}}|$, a correlation over coupled cells. The trust region is a live confound at large $|w_{\text{eff}}|$, the plug-in rule clipping on 884 sample steps where the contracting rung clips on 2,516, so a rung can be clip-limited and not controller-limited.

5 Discussion

Flow matching carries the study for its endpoint estimate and not a fidelity win, guidance works at a measured chemistry cost, and the ablations give one causal statement, that the dispersion term and only it produces the spread ladder, beside one null on coverage. Both transfer to the simplex.

Equation (5) explains the null: BDG is the plug-in field at a rescaled weight aiming at a shifted target, so it cannot exceed what that field reaches. The centring gain stays at w while the deviation gain is set online and may pass through zero, which is the one place BDG is ahead: more chemistry at matched coverage.

Three limitations bound the claim, each naming its next experiment: the trust region confounds the largest rungs, the signed- w control has not run, and the open-loop control rests on one seed pair. The narrowest claim supported is monotone, two-sided, modality-portable spread control at no extra sampling cost, which converts the coverage-against-fidelity trade-off into a quantity the user requests.

References

- [1] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” *Advances in neural information processing systems*, vol. 35, pp. 26565–26577, 2022.
- [2] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” *arXiv preprint arXiv:2210.02747*, 2022.
- [3] M. Albergo, N. M. Boffi, and E. Vanden-Eijnden, “Stochastic interpolants: A unifying framework for flows and diffusions,” *Journal of Machine Learning Research*, vol. 26, no. 209, pp. 1–80, 2025.
- [4] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [5] Q. Zheng, M. Le, N. Shaul, Y. Lipman, A. Grover, and R. T. Chen, “Guided flows for generative modeling and decision making,” *arXiv preprint arXiv:2311.13443*, 2023.
- [6] Authors listed on arXiv:2506.06085, “Feedback guidance of diffusion models,” *arXiv preprint arXiv:2506.06085*, 2025.
- [7] Authors listed on arXiv:2603.03281, “CFG-Ctrl: Classifier-free guidance as feedback control,” *arXiv preprint arXiv:2603.03281*, 2026.
- [8] G. Corso, Y. Xu, V. De Bortoli, R. Barzilay, and T. Jaakkola, “Particle guidance: Non-I.I.D. diverse sampling with diffusion models,” in *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.13102.
- [9] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” *arXiv preprint arXiv:2209.03003*, 2022.
- [10] N. Ma, M. Goldstein, M. S. Albergo, N. M. Boffi, E. Vanden-Eijnden, and S. Xie, “Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers,” in *European Conference on Computer Vision*, pp. 23–40, Springer, 2024.
- [11] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. C. Ye, “Diffusion posterior sampling for general noisy inverse problems,” in *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.14687.
- [12] J. Song, Q. Zhang, H. Yin, M. Mardani, M.-Y. Liu, J. Kautz, Y. Chen, and A. Vahdat, “Loss-guided diffusion models for plug-and-play controllable generation,” *Proceedings of the 40th International Conference on Machine Learning (PMLR v202)*, 2023. <https://proceedings.mlr.press/v202/song23k.html>.
- [13] B. Boys, M. Girolami, J. Pidstrigach, S. Reich, A. Mosca, and O. D. Akyildiz, “Tweedie moment projected diffusions for inverse problems,” *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2310.06721.
- [14] H. Ye, H. Lin, J. Han, M. Xu, S. Liu, Y. Liang, J. Ma, J. Zou, and S. Ermon, “Tfg: Unified training-free guidance for diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2409.15761.
- [15] H. Ben-Hamu, O. Puny, I. Gat, B. Karrer, U. Singer, and Y. Lipman, “D-flow: Differentiating through flows for controlled generation,” in *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.14017.
- [16] Authors listed on arXiv:2602.17211, “Moment guided diffusion,” *arXiv preprint arXiv:2602.17211*, 2026.
- [17] Authors listed on arXiv:2606.22239, “Variance-tilted diffusion,” *arXiv preprint arXiv:2606.22239*, 2026.
- [18] Authors listed on arXiv:2501.14822, “Controlling ensemble variance in diffusion models,” *arXiv preprint arXiv:2501.14822*, 2025.

- [19] E. Ventura, B. Achilli, L. Ambrogioni, and C. Lucibello, "Negative guidance reduces means and expands variances in diffusion models," *arXiv preprint arXiv:2602.00716*, 2026.
- [20] E. Hoogeboom, V. G. Satorras, C. Vignac, and M. Welling, "Equivariant diffusion for molecule generation in 3d," in *International conference on machine learning*, pp. 8867–8887, PMLR, 2022.
- [21] M. Xu, L. Yu, Y. Song, C. Shi, S. Ermon, and J. Tang, "Geodiff: A geometric diffusion model for molecular conformation generation," *arXiv preprint arXiv:2203.02923*, 2022.
- [22] R. Irwin, A. Tibo, J. P. Janet, and S. Olsson, "Semlaflow—efficient 3d molecular generation with latent attention and equivariant flow matching," *arXiv preprint arXiv:2406.07266*, 2024.
- [23] Y. Song, J. Gong, M. Xu, Z. Cao, Y. Lan, S. Ermon, H. Zhou, and W.-Y. Ma, "Equivariant flow matching with hybrid probability transport for 3d molecule generation," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [24] F. Bao, M. Zhao, Z. Hao, P. Li, C. Li, and J. Zhu, "Equivariant energy-guided SDE for inverse molecular design," *International Conference on Learning Representations (ICLR)*, 2023.
- [25] H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, and T. Jaakkola, "Dirichlet flow matching with applications to dna sequence design," *arXiv preprint arXiv:2402.05841*, 2024.
- [26] O. Davis, S. Kessler, M. Petrache, I. I. Ceylan, M. Bronstein, and A. J. Bose, "Fisher flow matching for generative modeling over discrete data," *Advances in Neural Information Processing Systems*, vol. 37, pp. 139054–139084, 2024.
- [27] S. Tang, Y. Zhang, A. Tong, and P. Chatterjee, "Gumbel-softmax flow matching with straight-through guidance for controllable biological sequence generation," *ArXiv*, pp. arXiv–2503, 2025.
- [28] T. Chen, Y. Zhang, S. Tang, and P. Chatterjee, "Multi-objective-guided discrete flow matching for controllable biological sequence design," *arXiv preprint arXiv:2505.07086*, 2025.

A Appendix

A.1 Dataset and Preprocessing Details

QM9 is the DeepChem packaging, pinned by SHA-256 on the archive, both source files and the built tensor. We keep the 3,054 uncharacterized molecules that EDM and its successors drop, a disclosed deviation. Splits come from one seeded permutation with $n_{\text{test}} = 13,083$ and $n_{\text{val}} = 17,748$, the remainder split in half. Normalisation constants from `train_a`: μ 2.70376 $\pm$ 1.53941 D, α 75.21779 $\pm$ 8.20041 Bohr³, gap 0.25126 $\pm$ 0.04746 Ha. The size histograms of the two training halves have total variation 0.0064 with mean atom counts 18.003 and 17.985, which makes them interchangeable as atom-count sources. Equivariance is verified numerically: 12 of 12 checks pass to $\sim 10^{-17}$ for rotation, translation, permutation and padding, on both the scalar and velocity heads. Modality 2 uses all 6,126 Kenyon-cell sequences with no train/validation split, because the evaluator is an exact count and no leakage is possible.

A.2 Training and Sampling Details

Guide and evaluator are E(3)-invariant networks of width 128 and 4 layers, 429,825 parameters, trained 120 epochs with Adam at 3×10^{-4} , cosine decay, batch 128, MSE on z -scored targets, selected on best validation MAE. Measured validation MAE on 17,748 molecules: f_A 0.0897 D, 0.2447, 0.00390 Ha; f_B 0.0840, 0.2407, 0.00380. Both are less accurate than published predictors because they are undertrained relative to them, and since δ scales with the evaluator’s MAE our bands are correspondingly wider, which is a disclosed conservatism. The sampler sweep at fixed checkpoint gives atom stability 0.9334, 0.9388, 0.9399, 0.9394 at NFE 100, 250, 500 and 1000, so quality saturates by 500 steps. Table 6 gives every trained component and Table 7 every sampler setting per arm.

Hardware and wall-clock for the sampling arms are recorded per cell in the run manifests and are summarised in Table 11.

Table 6: The three trained networks. Only the generator is large. Both predictors are frozen at sampling time and neither is updated by guidance. Values are read from the checkpoints’ own recorded arguments.

Setting	Generator v_θ	Guide f_A	Evaluator f_B
Role	samples	steers	scores
Used at sampling time	yes, 1 fwd/step	yes, 1 fwd + 1 bwd/guided step	no
Architecture	E(3)-equivariant EGNN	E(3)-invariant EGNN	E(3)-invariant EGNN
Width $\times$ depth	256×8	128×4	128×4
Parameters	3.75 M	429,825	429,825
Training split	<code>train_a</code> (51,527)	<code>train_a</code> (51,527)	<code>train_b</code> (51,527)
Objective	linear-interpolant FM	MSE on z -scored y	MSE on z -scored y
Optimiser	Adam	Adam	Adam
Learning rate	2×10^{-4} , cosine to 5%	3×10^{-4} , cosine	3×10^{-4} , cosine
Epochs / batch	1500 / 256	120 / 128	120 / 128
EMA decay	0.9999	none	none
Selection	final EMA weights	best validation MAE	best validation MAE
Seed	20260918	—	—
Wall-clock	11.4h (303k steps)	—	—
Checkpoint md5	a190ac83...	—	—

Table 7: Sampler settings per arm. Every arm shares one frozen generator, one frozen guide, one 100-step Euler grid and one trust region, so arms differ only in the guidance rule and its constants. A dash marks a constant the arm does not have. — marks a setting the v3 run fixes.

Arm	w	η	τ_{mult}	t_{min}	Batch B	Seeds	n per cell
unguided	—	—	—	—	256	1	256
plug-in	1	—	—	0.5	256	1	256
plug-in, strength sweep	4	—	—	0.5	256	1	256
BDG, $\eta=0$ gate	1	0	1.0	0.5	256	1	256
BDG, setpoint ladder	1	4	0.5–1.5	0.5	256	1	256
BDG, one-sided gate	1	4	1.5	0.5	256	1	256
BDG, high strength	32	4	0.5	0.5	256	1	256
v3 main run, all arms	—	—	—	0.5	—	3	5,000
Modality 2, all arms	—	—	—	—	—	—	—

A.3 Additional Ablations

Table 8 is the full five-rung setpoint ladder behind the three rungs quoted in the main text, on all three properties, with the controller’s own run-mean diagnostics beside the sample statistics. It is the measurement that the monotonicity claim rests on. Table 9 reserves the v3 extension of the same sweep.

Two facts in this table are worth separating. The spread column is monotone in τ_{mult} on all three properties, which is the control result. The IN-BAND column is not ordered and its best cell is the contraction rung, which also carries the lowest molecule stability on μ and gap, which is the fidelity cost. Neither the mean V_b/τ^2 nor w_{eff} settles at the equilibrium value $\tau^2(1 - 1/\eta)$ within the 50 guided steps, so these are transient, not converged, operating points.

A.4 Baseline Fidelity and Modality 2 Disclosures

Modality 1 naming. Our plug-in arm is a DPS-style Gaussian-energy field, where the original DPS differentiates the residual L_2 norm. Our TMPD arm replaces the constant denominator by $s^2 + g^\top \Sigma g$, which is the published method’s scalar affine-property limit, included as a related-work comparison rather than a faithful nonlinear port. TFG runs at its published QM9 hyperparameters through our deterministic ODE sampler where the original used a stochastic DDIM update, and our unscaled one-hot features move roughly 16 times less relative to coordinates than the released checkpoint’s, so its numbers here must never be read against its published table.

Modality 2 deviations. The standard enhancer protocol is class-conditional generation at 500 base pairs on 81-class or 47-class datasets of roughly 90,000 sequences, scored by Fréchet biological distance over 10,000 samples per model. We use GC-band coverage instead of class-conditional

Table 8: The complete setpoint ladder at $\eta=4$, $w=1$, q50 target, one seed, $n=256$ per cell. Spread is $\text{sd}(f_A)$ over the batch divided by the unguided arm’s. V_b/τ^2 and w_{eff} are run means over guided steps. The two gate rows are exact: $\eta=0$ is bit-identical to the plug-in arm on all three properties, and the one-sided variant at an expanding setpoint holds $w_{\text{eff}}=+1.000$. **Resolution:** at $n=256$ the standard error on IN-BAND is about 0.019 near $p=0.1$, so IN-BAND differences below roughly two points are not resolvable here, and the BDG rows’ error is understated because the batch is coupled through $\bar{F}$ and V_b . The ordering of the spread column, not the size of any single IN-BAND difference, is what this table establishes.

Property	Arm	τ_{mult}	Spread / ung.	V_b/τ^2	w_{eff}	IN-BAND $\uparrow$	Mol. stab. $\uparrow$
μ	unguided	—	1.000	—	—	0.0898	0.4336
μ	plug-in	—	0.714	—	+1.000	0.1094	0.3633
μ	BDG	0.50	0.670	1.2731	+2.093	0.1406	0.3359
μ	BDG	0.75	0.813	0.9351	+0.741	0.0820	0.4102
μ	BDG	1.00	0.900	0.7578	+0.031	0.0820	0.4258
μ	BDG	1.25	0.982	0.6746	−0.302	0.0742	0.4219
μ	BDG	1.50	1.093	0.6459	−0.416	0.0703	0.4141
α	unguided	—	1.000	—	—	0.0508	0.4336
α	plug-in	—	0.849	—	+1.000	0.0547	0.3711
α	BDG	0.50	0.674	1.5365	+3.146	0.0742	0.3711
α	BDG	0.75	0.844	1.0761	+1.305	0.0547	0.3633
α	BDG	1.00	1.014	0.8777	+0.511	0.0625	0.3711
α	BDG	1.25	1.099	0.7190	−0.124	0.0547	0.3672
α	BDG	1.50	1.187	0.6166	−0.533	0.0430	0.3945
gap	unguided	—	1.000	—	—	0.1211	0.4336
gap	plug-in	—	0.871	—	+1.000	0.1562	0.3359
gap	BDG	0.50	0.742	1.3621	+2.448	0.1406	0.3086
gap	BDG	0.75	0.893	1.0024	+1.009	0.1055	0.3672
gap	BDG	1.00	0.968	0.7781	+0.113	0.0977	0.3906
gap	BDG	1.25	1.052	0.6376	−0.450	0.1016	0.3594
gap	BDG	1.50	1.073	0.5401	−0.840	0.0938	0.3203

Table 9: Reserved: the v3 extension of the ablation, three seeds and two backends at $n=5,000$. Every cell reports IN-BAND with its seed standard deviation and molecule stability. Row and column structure, the grid and the ordering are final; only the values are pending. This grid exists to separate the two confounds the one-seed table above cannot: whether monotonicity survives seed noise, and whether the gain η and the nominal strength w act independently.

η (at $w=1$)	τ_{mult}				
	0.50	0.75	1.00	1.25	1.50
0	—	—	—	—	—
1	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—
8	—	—	—	—	—
η (at $w=4$)	0.50	0.75	1.00	1.25	1.50
0	—	—	—	—	—
1	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—
8	—	—	—	—	—

distance, 6,126 sequences instead of 83,968, and 200 base pairs in the shipped cells. Three further facts bear on the comparison. Gumbel-Softmax flow matching does not run the enhancer benchmark at all, reporting promoter, protein and peptide tasks, so only two of the three closest named methods apply here. Fisher Flow disputes the standard evaluator, reporting that the cell-type classifiers behind it score 11.5% and 11.2% test accuracy. MOG-DFM’s own guided demonstration runs at length 100 while reporting base distances at 500, so short-length guided evaluation is precedented. At 200 base pairs the reference classifier global-average-pools and returns meaningless numbers rather than failing loudly, which is a further reason we do not quote it.

Per-seed values. Table 10 reserves the per-seed breakdown behind every aggregate in the main text. Three seeds is the minimum at which a standard deviation means anything, and with $n=5,000$ per cell the standard error on IN-BAND near $p=0.1$ is about 0.004, roughly five times finer than the one-seed ablation above. Seed-to-seed spread is reported for every arm, including the arms whose aggregates

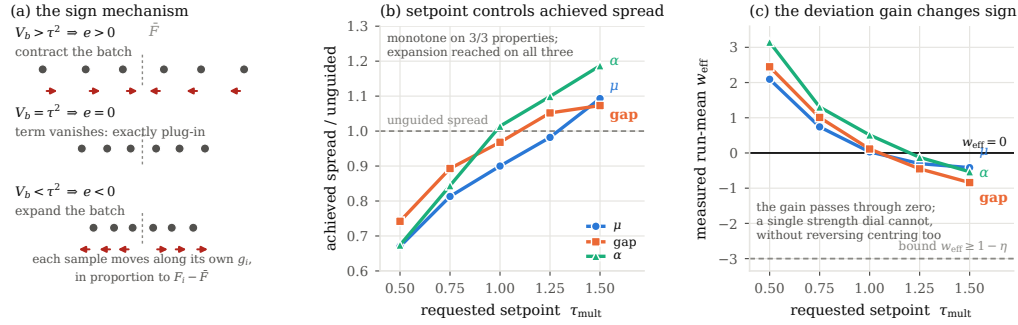


Figure 2: The controller, from the measurements in Table 8. Left: the rule. The deviation gain is servoed while the centring gain is held at w , so the step is the plug-in field at a rescaled weight aiming at a shifted target. Middle: the setpoint ladder. Requested spread against achieved spread on all three properties, monotone on all of them, with the unit line marking the unguided spread. Right: the signed effective gain w_{eff} along the ladder. It crosses zero, which is what makes the control two-sided: the same rule contracts and expands depending on the sign of the dispersion error.

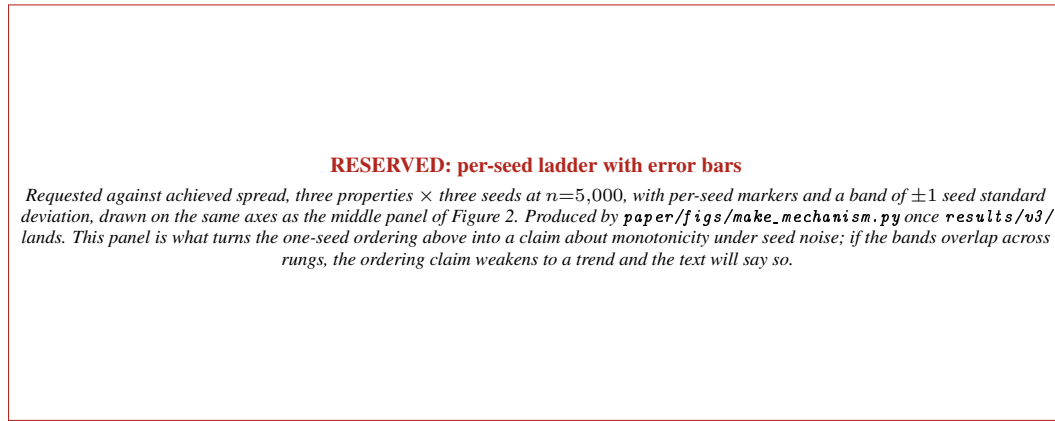


Figure 3: Reserved: monotonicity under seed noise.

we do not claim a difference for, so that a reader can check whether a difference we do claim exceeds it.

Table 10: Reserved: per-seed IN-BAND and molecule stability for every arm in Tables 2 to 4 of the main text, $n=5,000$ per cell, q50 target. Rows, columns and the seeds are final; only the values are pending. The last column is the quantity every main-text comparison must clear.

Arm	Property	Seed 1	Seed 2	Seed 3	Mean	Seed sd
unguided	$\mu / \alpha / \text{gap}$	—	—	—	—	—
plug-in	$\mu / \alpha / \text{gap}$	—	—	—	—	—
TMPD-inspired	$\mu / \alpha / \text{gap}$	—	—	—	—	—
LGD-MC	$\mu / \alpha / \text{gap}$	—	—	—	—	—
TFG	$\mu / \alpha / \text{gap}$	—	—	—	—	—
D-Flow	$\mu / \alpha / \text{gap}$	—	—	—	—	—
BDG (ours)	$\mu / \alpha / \text{gap}$	—	—	—	—	—

A.5 Additional Samples and Failure Cases

Figures 4 and 5 are the qualitative counterpart to Table 8. Their purpose is adversarial: the contraction rung is the cell with the best IN-BAND and the worst molecule stability, so it is the cell whose samples most need to be shown rather than summarised.

RESERVED: molecule grid across the setpoint ladder

A 5×8 grid of decoded molecules, one row per ladder rung $\tau_{\text{mult}} \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ at $\eta=4$, $w=1$, on μ , drawn from a fixed seed so rows are paired sample for sample. Each molecule is annotated with its f_B property value, whether it falls in the band, and whether it passes the EDM stability check. Unstable molecules are kept in the grid and marked, not filtered out.

Figure 4: Reserved: samples along the setpoint ladder.

RESERVED: failure cases at the contraction rung

The eight lowest-stability molecules from the $\tau_{\text{mult}}=0.5$ cell beside their paired unguided samples from the same seed and noise draw, with the per-atom valency violations marked. Below them, eight generated DNA sequences at the matching contraction rung with their GC content and the band edges. This panel exists to show what the fidelity cost in Table 8 actually looks like.

Figure 5: Reserved: what the contraction rung breaks.

A.6 Algorithms

Algorithm 1 One guided sampling step with Batch-Dispersion Guidance. Nothing is trained at sampling time; the generator v_θ and guide f_A are frozen. Modality-specific inputs are the endpoint map and the property f_A .

Require: batch $x_t \in \mathbb{R}^{B \times d}$, time t , strength w , gain η , setpoint τ , scale s , step h

- 1: $v_i \leftarrow v_\theta(x_t^{(i)}, t)$ ▷ one generator pass, shared with the unguided sampler
- 2: **if** $t < t_{\min}$ **then return** $x_t + h v$ ▷ guidance window
- 3: **end if**
- 4: $m_i \leftarrow x_t^{(i)} + (1 - t) v_i$ ▷ endpoint estimate
- 5: $F_i \leftarrow f_A(m_i)$; $g_i \leftarrow \nabla_m f_A(m_i)$ ▷ one guide forward, one backward
- 6: $\bar{F} \leftarrow \frac{1}{B} \sum_i F_i$; $V_b \leftarrow \frac{1}{B-1} \sum_i (F_i - \bar{F})^2$ ▷ the batch is the estimator
- 7: $e \leftarrow (V_b - \tau^2)/\tau^2$; **if** one-sided **then** $e \leftarrow \max(e, 0)$
- 8: $\text{num}_i \leftarrow (y - F_i) - \eta e (F_i - \bar{F})$ ▷ $= (y - \bar{F}) - w_{\text{eff}}(F_i - \bar{F})$, $w_{\text{eff}} = 1 + \eta e$
- 9: $G_i \leftarrow (\text{num}_i/s^2) J_i^\top g_i$ ▷ one vector-Jacobian product
- 10: $G_i \leftarrow w \frac{1-t}{t} G_i$; $G_i \leftarrow G_i \cdot \min(1, \|v_i\|/\|G_i\|)$ ▷ trust region, shared by all arms
- 11: **return** $x_t + h(v + G)$

If $B < 2$ the unbiased variance is undefined, and the implementation sets $V_b := \tau^2$ so that $e = 0$ and the step degrades to the plug-in rule rather than emitting a non-finite value.

A.7 Implementation and Reproducibility Details

Every number in this paper is produced by one script reading one recorded artefact, listed in Table 11. The innovation itself is the `mode == "bdg"` branch of `proj1/src/guidance.py`, gated by 25 checks in `proj1/tests/test_bdg.py`, which include the two exactness properties asserted in Table 8.

Table 11: Provenance. Each row names the object in the paper, the script that emits it and the recorded artefact that script reads. `—` marks an artefact the v3 run has not written. Nothing in the paper is assembled by hand.

Object	Produced by	Reads
Table 1, base models	<code>scripts/benchmark_base.py</code>	<code>weights/fm_ema.pt, results/bench/</code>
Table 2, guidance arms	<code>scripts/onegen_full_metrics.py</code>	<code>results/bdg_local/</code>
Table 3, BDG ablation	<code>scripts/bdg_full_metrics.py</code>	<code>results/bdg_port/cells/</code>
Table 4, recent methods	<code>scripts/transfer_sweep.py</code>	<code>results/v3/ —</code>
Table 5, Modality 2	<code>m2/m2_sweep.py</code>	<code>results/m2/ —</code>
Table 8	<code>scripts/bdg_table.py</code>	<code>results/bdg_port/table.txt</code>
Table 9	<code>scripts/bdg_table.py</code>	<code>results/v3/ —</code>
Figure 1, overview	<code>paper/figs/overview.tex</code>	nothing; it is a diagram
Figure 2	<code>paper/figs/make_mechanism.py</code>	<code>results/bdg_port/table.txt</code>
Figures 3–5	<code>paper/figs/make_mechanism.py</code>	<code>results/v3/ —</code>
The innovation	<code>proj1/src/guidance.py, mode == "bdg"</code>	—
Its gates	<code>proj1/tests/test_bdg.py (25 checks)</code>	—

Versions and identifiers. The generator checkpoint is pinned by md5 `a190ac83...` and QM9 by SHA-256 on both the source archive and the built tensor. Software versions, the remaining checkpoint hashes and the per-cell run manifests are `—` until the v3 environment is frozen; they are recorded by the same scripts that emit the tables above, so they land with the numbers rather than being transcribed after the fact.